LAB ASSIGNMENT 7

TEAM MEMBER DETAILS

MEMBER_1 :KARAN (LEADER)

COLLEGE_ID :20261651048

MEMBER_2 :HIMANSHU BAJPAI

COLLEGE_ID :20261651041

MEMBER_3 :DIVYANSHU TIWARI

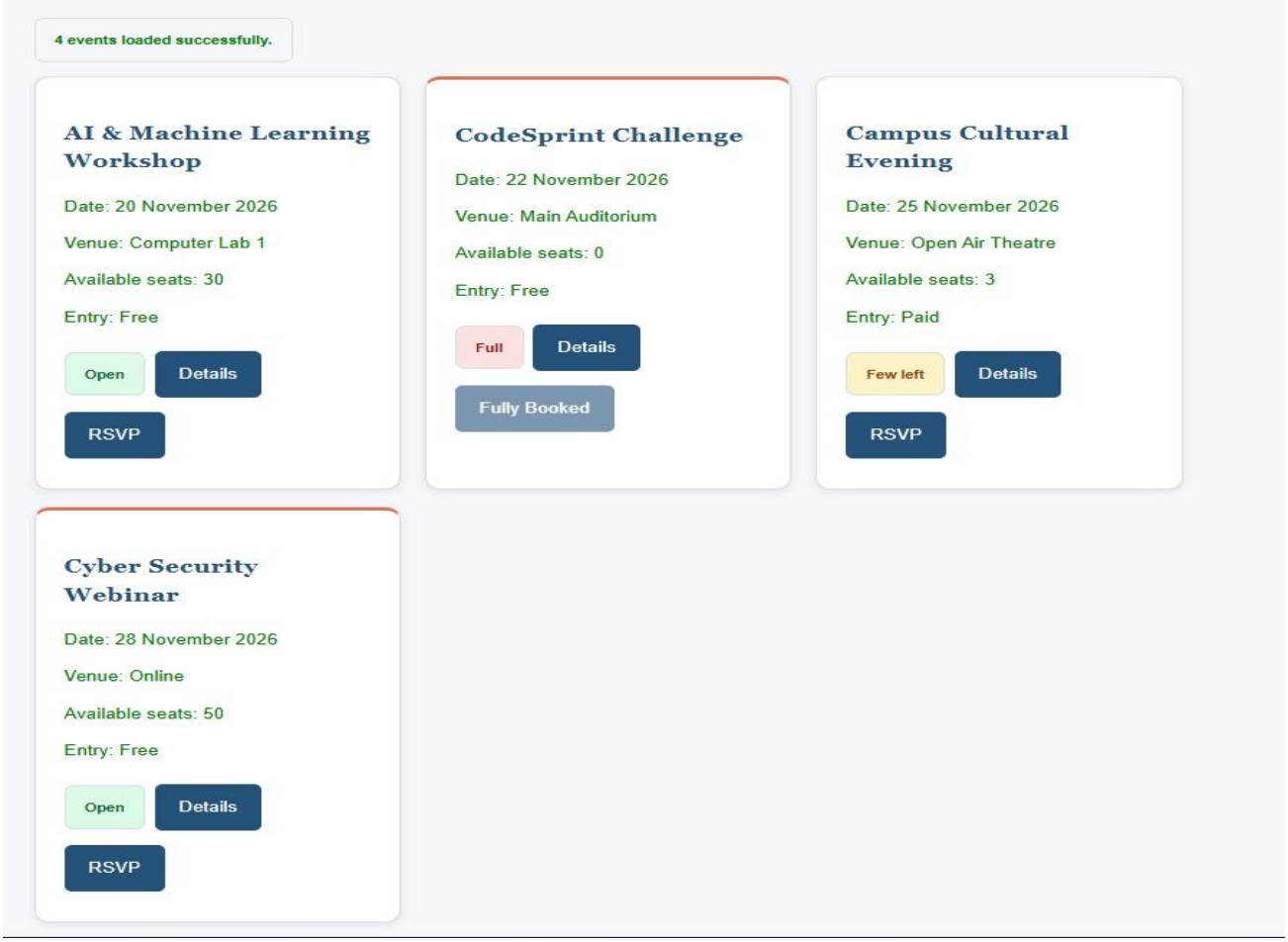
COLLEGE_ID :20261651034

MEMBER_4 : ETNI HARSHITHA

COLLEGE_ID :20261651036

Screenshots

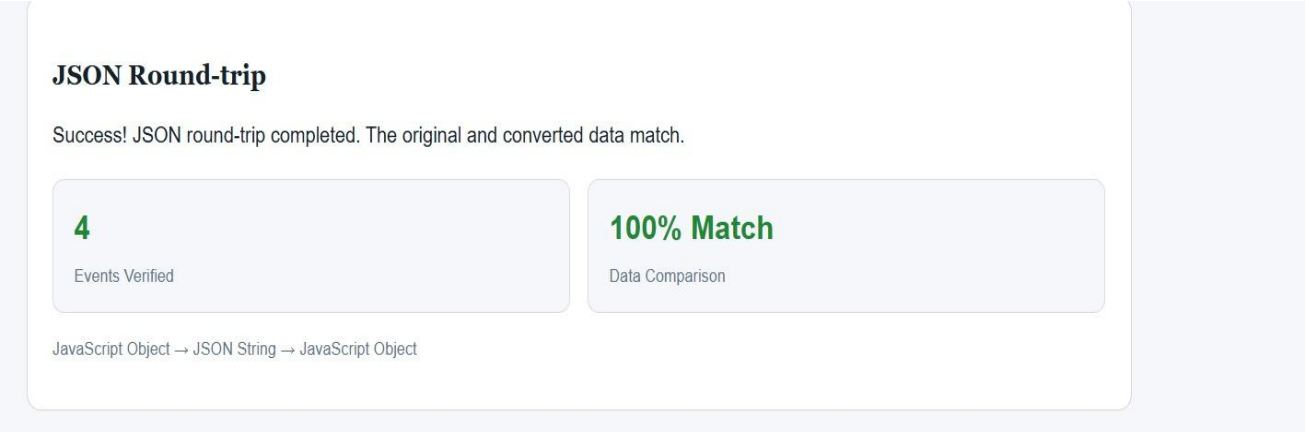
Screenshot 1 — JSON Ev ents Display



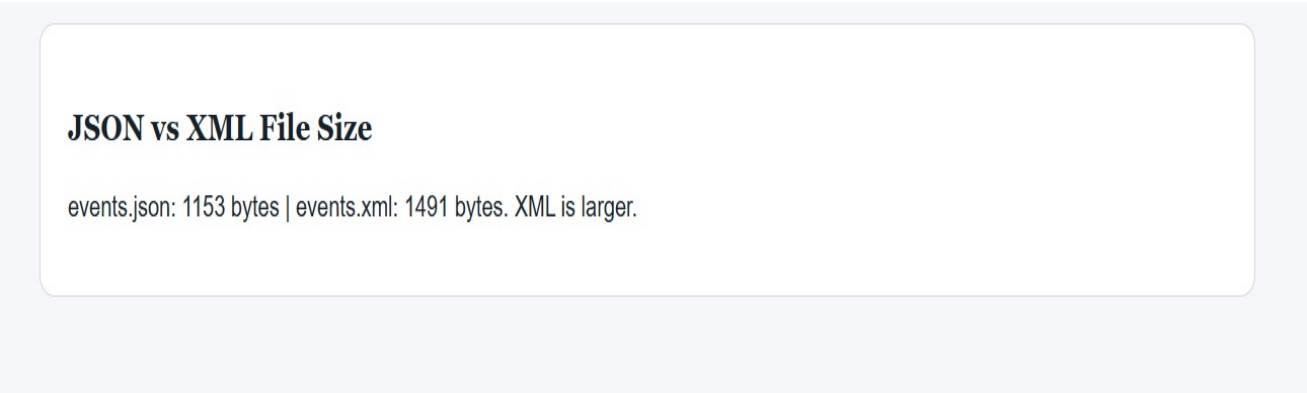
Screenshot 2 — XML Data Display



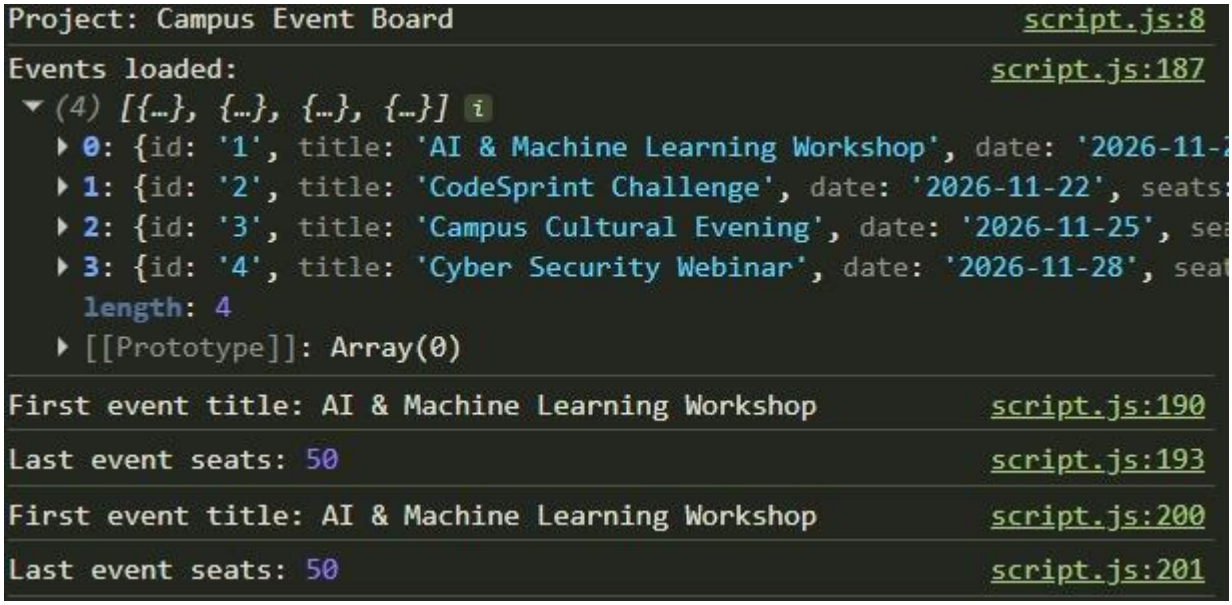
Screenshot 3 — JSON Round-trip



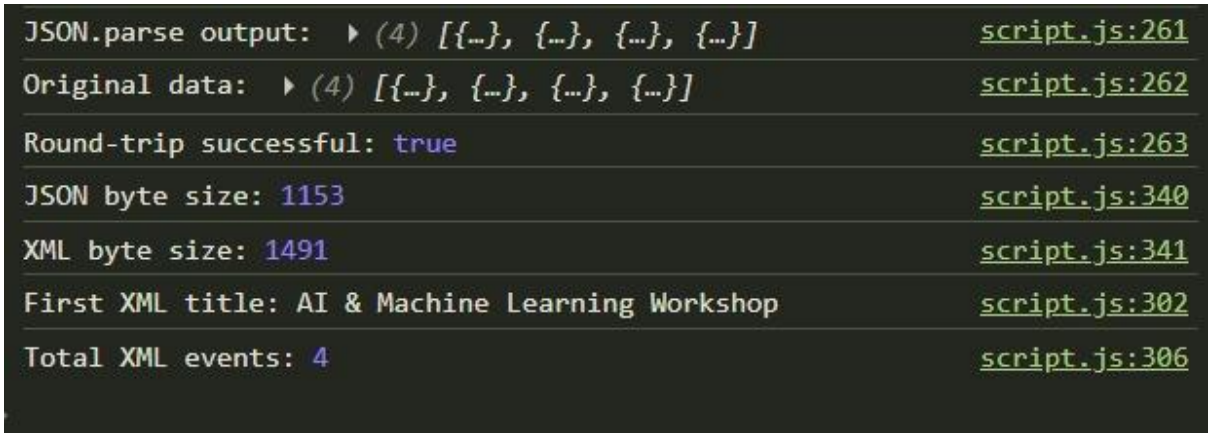
Screenshot 4 — File Size Comparison



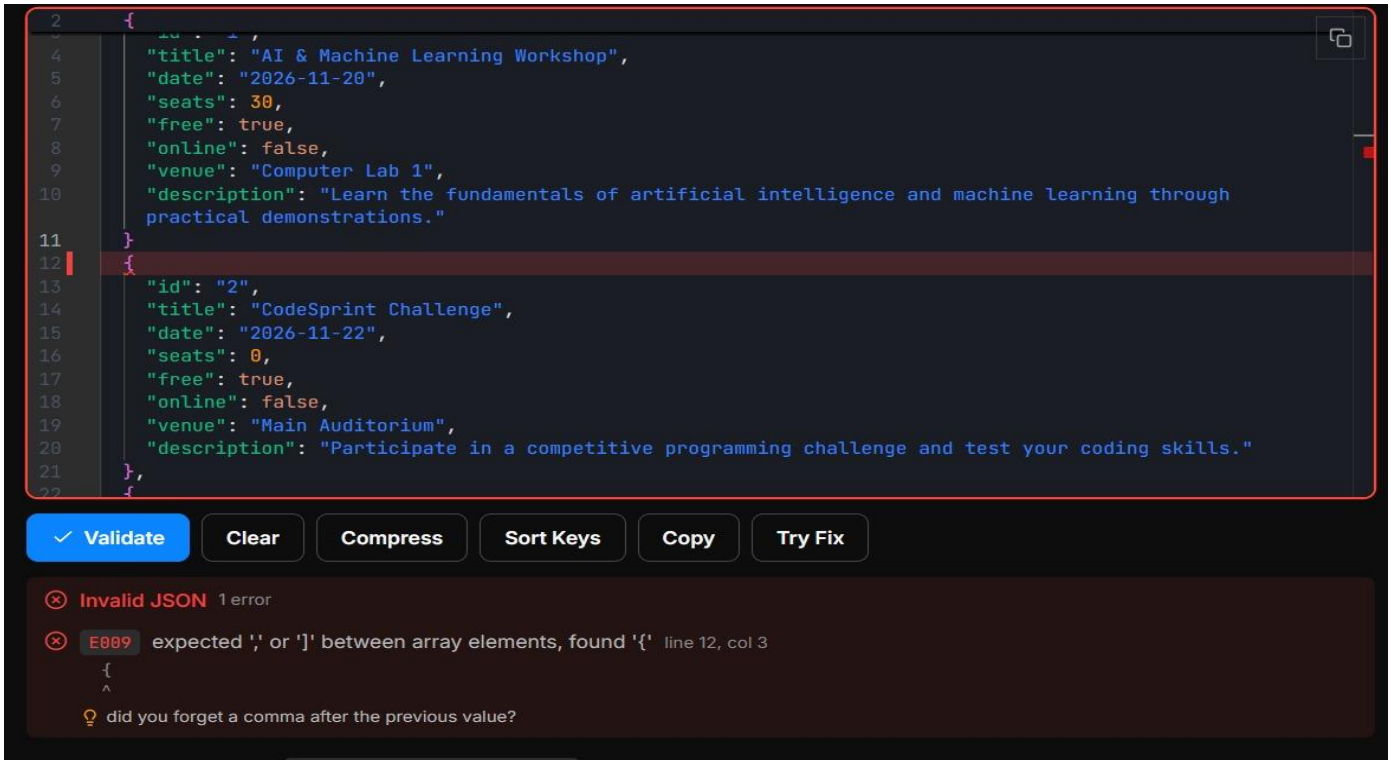
Screenshot 5 — JSON Console Output



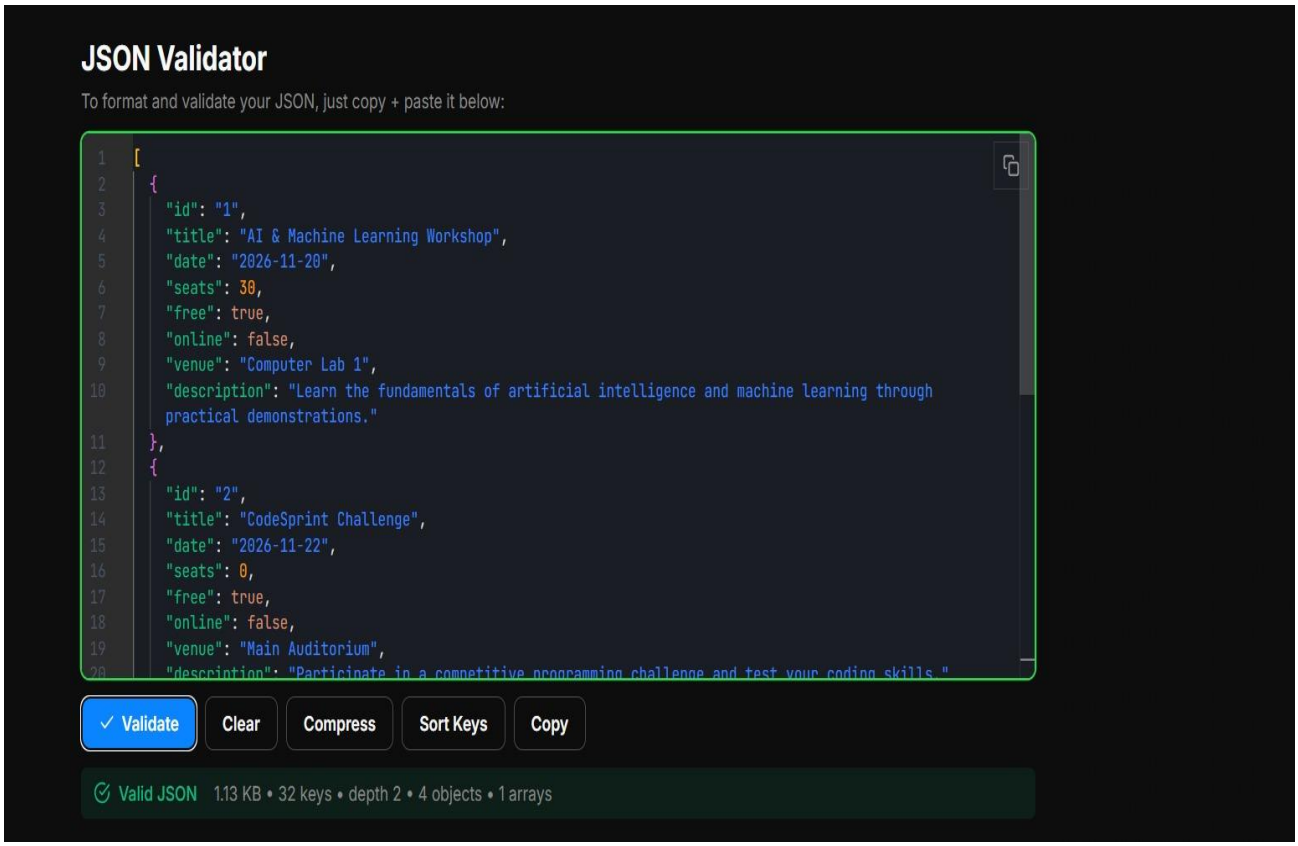
Screenshot 6 — XML Console Output



Screenshot 7 — Invalid JSON



Screenshot 8 — Corrected JSON



Question 1. Show your event table, then one event as JSON and XML side by side. Point out three correspondences.

Event Table

ID	Title	Date	Seats	Free
1	AI & Machine Learning Workshop	2026-11-20	30	true
2	CodeSprint Challenge	2026-11-22	0	true
3	Campus Cultural Evening	2026-11-25	3	false
4	Cyber Security Webinar	2026-11-28	50	true

JSON Representation

```
{
  "id": "1",
  "title": "AI & Machine Learning Workshop",
  "date": "2026-11-20",
  "seats": 30,
  "free": true
}
```

XML Representation

```
<?xml version="1.0" encoding="UTF-8"?>
<events>
    <event id="1">
        <title>AI &amp;amp; Machine Learning Workshop</title>
        <date>2026-11-20</date>
        <seats>30</seats>
        <free>true</free>
    </event>
</events>
```

Three Correspondences

1. Table column `title` corresponds to JSON key `"title"` and XML element `<title>`.
2. Table column `date` corresponds to JSON key `"date"` and XML element `<date>`.
3. Table column `seats` corresponds to JSON key `"seats"` and XML element `<seats>`.

Question 2. Which JSON value types did you use? Give one value of each you used.

I used three main JSON value types:

1. **String:** `"AI & Machine Learning Workshop"`

2. **Number:** `30`
3. **Boolean:** `true`

A string stores text, a number stores numeric values, and a boolean stores either `true` or `false`.

Question 3. Why must a JSON key be in double quotes? Write one line that is valid JavaScript but invalid JSON.

JSON requires all object keys to be strings enclosed in double quotation marks. This is part of the JSON syntax.

Valid JSON:

```
{"title": "AI Workshop"}
```

Valid JavaScript but invalid JSON:

```
{title: "AI Workshop"}
```

The second example is invalid JSON because the key `title` is not enclosed in double quotation marks.

Question 4. In events.xml, which field is an attribute and which are elements? Justify element vs attribute.

In my `events.xml` file, `id` is an attribute of the `<event>` element.

Example:

```
<event id="1">
    <title>AI &amp;amp; Machine Learning Workshop</title>
    <date>2026-11-20</date>
    <seats>30</seats>
    <free>true</free>
</event>
```

The fields `title`, `date`, `seats`, `free`, `online`, `venue`, and `description` are child elements.

I used an attribute for `id` because it identifies each event. The other fields are elements because they contain the actual event information.

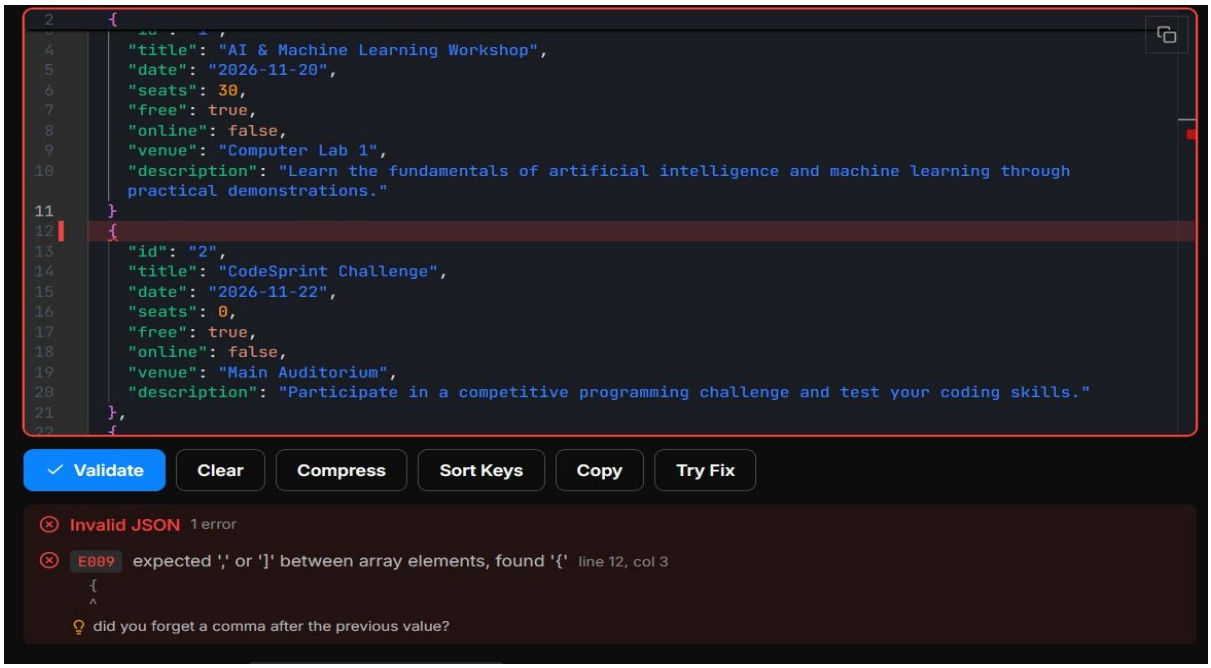
Question 5. Paste the validator error from your broken JSON (C1) and the fix that made it valid again.

I intentionally introduced a trailing comma after the last property in my JSON object.

Incorrect JSON

```
{
    "id": "1",
    "title": "AI Workshop",
    "seats": 30,
}
```

Validator Error



Corrected JSON

Question 6. How did you read a single value in JSON (dot/index) and in XML (getElementsByTagName)? Paste both lines.

Reading JSON

I used array index access and dot notation to read event properties.

```
console.log("First event title:", events[0].title);
```

```
console.log(
    "Last event seats:",
    events[events.length - 1].seats
);
```

The first line accesses the title of the first event. The second line accesses the seats of the last event.

Reading XML

I used `getElementsByTagName()` to access XML elements.

```
const titles = xmlDocument.getElementsByTagName("title");

const firstTitle = titles[0].textContent;
```

This retrieves the title of the first event from the XML document.

Question 7. Report the byte sizes of events.json and events.xml. Why is the XML larger?

The file sizes measured in my project are:

File	Size
events.json	1,153 bytes
events.xml	1,491 bytes

Difference:

1491 - 1153 = 338 bytes.

Therefore, XML is 338 bytes larger than JSON in my project.

XML is larger because it uses opening and closing tags for each element, along with additional markup. JSON uses compact key-value pairs and generally requires less structural text for this data.

Question 8. Neither format has a real date type — what convention did you use, and why is it unambiguous worldwide?

I used the ISO-8601 date format:

```
YYYY-MM-DD
```

Example:
`2026-11-20`

This format stores the year first, followed by the month and day. It avoids confusion between different regional date formats, such as DD-MM-YYYY and MM-DD-YYYY.

Both JSON and XML store these dates as strings in my project.

Question 9. List your 3 schema rules (C2). Why is it important to validate data before your code uses it?

My three data validation rules are:

1. Every event must have a non-empty title.
2. The seats value must be a non-negative integer.
3. Every event date must follow the ISO-8601 `YYYY-MM-DD` format.

Validation is important because incorrect or missing data can cause application errors. It ensures that the data follows the expected structure and format before JavaScript uses it.

These rules represent the basic idea of a data schema.

Question 10. Which format did you choose for the page, and why? Give one scenario where XML is the better choice. Paste the GitHub repo link.

I chose **JSON** as the main data format for my Campus Event Board.

Reasons:

1. JavaScript can directly convert JSON into objects and arrays.
2. JSON is easy to read and write.
3. It works conveniently with the Fetch API.
4. It requires less structural markup than XML for my event data.

When XML is a better choice

XML is useful when exchanging complex, structured documents between enterprise systems, especially when those systems use XML-based standards and require attributes, namespaces or document validation.

GitHub Repository

https://github.com/karanbaira5/event-board